



Simulation of MOSFET Using SILVACO Tool

Major Task Part 2

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Chapter 1

Introduction

1.1 Physics and Electronics Background

The Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is the fundamental building block of modern solid-state electronics, serving as the primary switching and amplification device in both digital and analog integrated circuits. Its operation is governed by the field effect: a voltage applied to an insulated gate terminal modulates the inversion charge density at the oxide-semiconductor interface, thereby controlling the conductivity of a semiconductor channel situated between the source and drain terminals.

For an N-channel MOSFET (NMOS), which is the focus of this study, the device is constructed using a p-type silicon substrate with heavily doped n-type regions for the source and drain. When a positive gate-to-source voltage (V_{GS}) exceeding the threshold voltage (V_{th}) is applied, the resulting vertical electric field repels majority-carrier holes from the surface and attracts minority-carrier electrons to the Si/SiO₂ interface. This phenomenon, known as strong inversion, forms an n-type conductive channel connecting the source and drain, allowing a drain current (I_D) to flow when a drain-to-source voltage (V_{DS}) is applied. A schematic representation of this internal structure alongside its standard circuit symbol is illustrated in Figure 1.1. This report focuses on accurately modeling this behavior, including secondary effects such as channel length modulation, body effect, and mobility degradation, all of which are directly characterized through Silvaco Atlas TCAD simulation and SPICE parameter extraction.

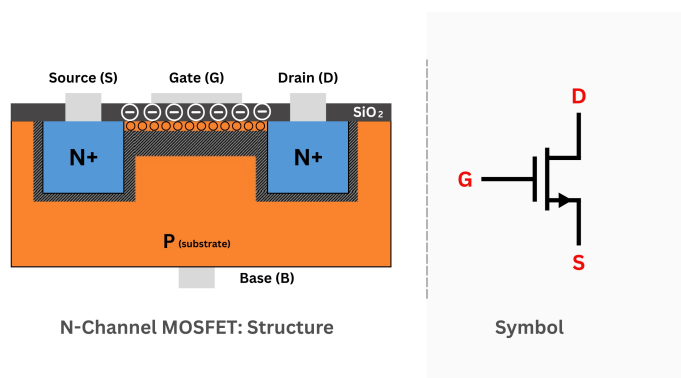


Figure 1.1: Cross-sectional structure of an N-channel MOSFET showing the inversion layer channel (left) and its corresponding schematic symbol (right).

1.2 Simulation Environment

To accurately model the semiconductor physics and extract the SPICE parameters, this project employs Silvaco TCAD (Technology Computer-Aided Design), specifically the Atlas device simulation framework. The simulations were executed using the nanoHUB platform, running within a Linux Virtual Machine environment. Silvaco Atlas utilizes numerical, grid-based methods to solve the fundamental semiconductor transport equations — Poisson’s equation and the electron and hole continuity equations — thereby simulating the electrical characteristics of the defined MOSFET structure under various DC and small-signal AC bias conditions.

Two physical models were enabled in all simulations. The Lombardi CVT (Concentration and Temperature dependent) mobility model accounts for surface roughness scattering, acoustic phonon scattering, and ionized impurity scattering at the Si/SiO₂ interface, consistent with the effective mobility degradation framework covered in the course. The Shockley-Read-Hall (SRH) model accounts for trap-assisted generation and recombination through mid-gap interface states.

1.3 Device Specifications and Initial Parameters

The simulated device is a 2D planar N-channel MOSFET. The physical structure, mesh definitions, and doping concentrations were defined in the Atlas syntax. The physical dimensions and initial device parameters are summarized in Table 1.1.

Table 1.1: MOSFET Physical Dimensions and Initial Parameters

Parameter	Symbol / Region	Value
Total Device X-Extent (source pad + channel + drain pad)	L_{total}	1.4 μm
Channel Length	L	1.0 μm
Channel Width (2D unit depth, z-direction)	W	1.0 μm
Gate Oxide Thickness	t_{ox}	0.02 μm (20 nm)
Substrate Depth	y_{sub}	0.3 μm
Substrate Doping	N_A (P-type, Uniform)	$1 \times 10^{16} \text{ cm}^{-3}$
Source/Drain Doping	N_D (N-type, Uniform)	$1 \times 10^{20} \text{ cm}^{-3}$
Gate Material	—	Ideal Conductor (Assumed)
Dielectric Material	—	Silicon Dioxide (SiO ₂)
Substrate Material	—	Silicon (Si)
Physical Models	—	CVT (Mobility), SRH

**The ideal conductor gate assumption sets the metal–semiconductor work function difference $\Phi_{ms} = 0$, which eliminates the flatband voltage contribution $V_{FB} = \Phi_{ms} - Q_{oi}/C_{ox}$ from the threshold voltage expression $V_{th} = V_{FB} + 2\phi_F - Q_d/C_{ox}$. Consequently, the simulated threshold voltage V_{TO} will differ from an experimentally fabricated device with a polysilicon or metal gate. This assumption is acknowledged as a limitation of the present simulation.*

The corresponding 2D structural profile and absolute net doping concentration are visualized in Figure 1.2. The plot highlights the clear physical boundaries between the heavily doped n^+ source and drain regions ($1 \times 10^{20} \text{ cm}^{-3}$) and the moderately doped p -type bulk silicon substrate ($1 \times 10^{16} \text{ cm}^{-3}$). The thin SiO_2 dielectric layer is aligned atop the channel region between $x = 0.2 \text{ }\mu\text{m}$ and $x = 1.2 \text{ }\mu\text{m}$, isolating the gate electrode as defined by the structural mesh. This figure corresponds to the initial unbiased structure ($V_{GS} = V_{DS} = V_{BS} = 0 \text{ V}$).

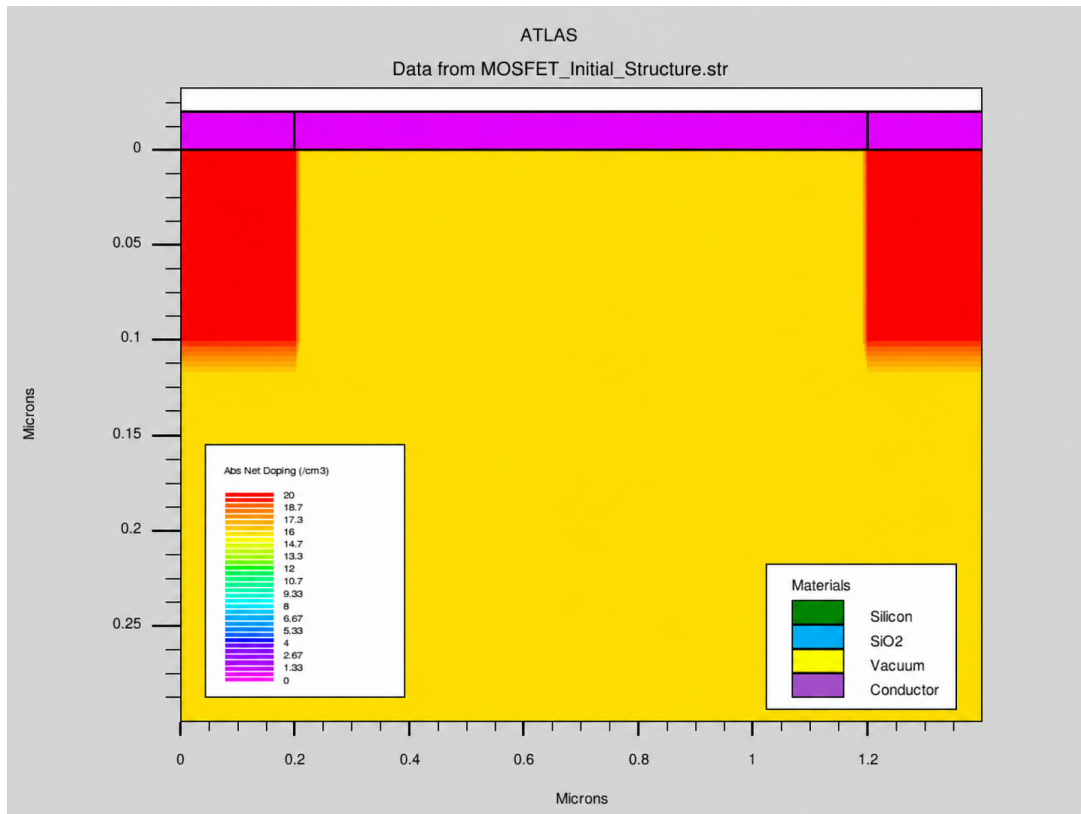


Figure 1.2: Silvaco Tonyplot 2D structure showing the absolute net doping concentration profile ($\log_{10}$ scale, cm^{-3}) and material regions of the simulated N-channel MOSFET under zero-bias initial conditions ($V_{GS} = V_{DS} = V_{BS} = 0 \text{ V}$). Red regions: heavily doped n^+ source/drain ($N_D = 1 \times 10^{20} \text{ cm}^{-3}$); yellow region: p-type silicon substrate ($N_A = 1 \times 10^{16} \text{ cm}^{-3}$); green layer: SiO_2 gate dielectric ($t_{ox} = 20 \text{ nm}$); purple layer: ideal conductor gate electrode.

Chapter 2

Simulation Results and Discussion

This chapter presents the electrical characterization of the simulated $1.0\ \mu\text{m}$ N-channel MOSFET. The evaluation begins with the external terminal DC behavior via transfer characteristics and output characteristics, followed by body effect analysis. Internal device physics are then examined through electrostatic potential profiles, energy band diagrams, and carrier concentration maps. Finally, the small-signal AC capacitance characteristic is evaluated. Together, these results provide the data used to extract the primary SPICE model parameters presented in Chapter 3.

2.1 DC Transfer Characteristics (I_D – V_{GS})

The switching behavior of the simulated NMOS is initially evaluated through its linear-scale transfer characteristics. Figure 2.1 shows the drain current (I_D) as a function of gate-to-source voltage (V_{GS}) swept from 0 V to 5 V, at four drain bias conditions ($V_{DS} = 0.1\ \text{V}$, $1.0\ \text{V}$, $2.0\ \text{V}$, and $3.0\ \text{V}$).

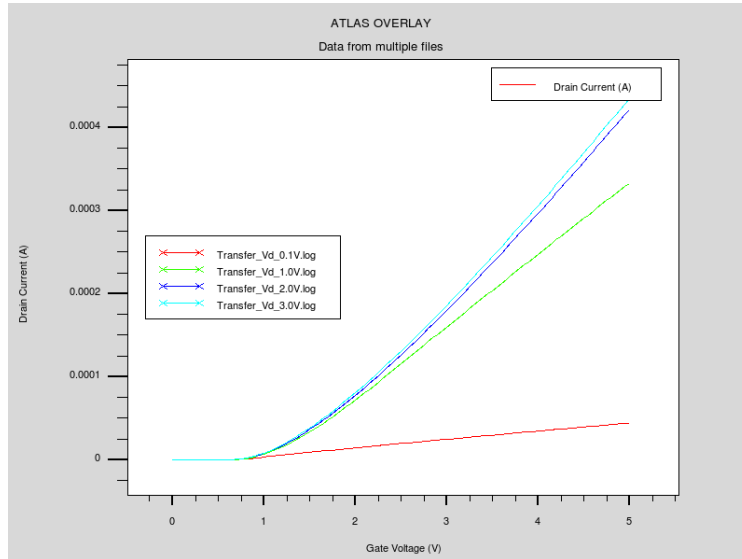


Figure 2.1: Linear-scale DC transfer characteristics: drain current I_D (A) vs. gate-to-source voltage V_{GS} (V) for the $1.0\ \mu\text{m}$ NMOS at $V_{DS} = 0.1\ \text{V}$ (red), $1.0\ \text{V}$ (green), $2.0\ \text{V}$ (blue), and $3.0\ \text{V}$ (cyan). Substrate and source grounded ($V_{BS} = 0\ \text{V}$).

Physical and Operational Analysis

1. **Subthreshold and Cutoff Regime** ($V_{GS} < V_{th}$): At gate voltages below the threshold voltage ($V_{th} \approx 0.676$ V, as extracted in Chapter 3), the drain current is negligible on a linear scale across all V_{DS} curves. In this state, the surface potential has not yet reached the strong inversion condition ($\phi_s < 2\phi_F$), and the energy barrier between the source and the channel prevents significant carrier injection. The device remains in its OFF state.
2. **Linear (Triode) Region** ($V_{DS} = 0.1$ V): For the low drain bias condition (red curve, $V_{DS} = 0.1$ V), the longitudinal electric field along the channel is weak and uniform. As V_{GS} surpasses the threshold voltage, the device enters the triode region. Using the simplified long-channel approximation (valid when $V_{DS}^2/2 \ll (V_{GS} - V_{th})V_{DS}$), the drain current reduces to:

$$I_D \approx \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2} \right] \approx \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})V_{DS} \quad (2.1)$$

The resulting linear I_D - V_{GS} relationship at $V_{DS} = 0.1$ V is used in the Atlas extraction script to determine the threshold voltage V_{TO} via maximum-slope linear extrapolation, and to extract the transconductance parameter KP .

3. **Saturation Region** ($V_{DS} \geq 1.0$ V): As V_{DS} increases to 1.0 V, 2.0 V, and 3.0 V, the inversion layer experiences pinch-off at the drain end once $V_{GD} < V_{th}$ (i.e., $V_{DS} \geq V_{GS} - V_{th}$). The drain current transitions from a linear to a quadratic dependence on the gate overdrive voltage:

$$I_D \approx \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS}) \quad (2.2)$$

This is the standard long-channel saturation model taught in the course, where λ is the channel length modulation parameter extracted in Chapter 3.

4. **Channel Length Modulation Observation:** The gap between the $V_{DS} = 2.0$ V and $V_{DS} = 3.0$ V curves is noticeably smaller than the gap between $V_{DS} = 0.1$ V and $V_{DS} = 1.0$ V. Once in saturation, the drain current is largely independent of further increases in V_{DS} . The minor upward shift is due to channel length modulation ($\lambda > 0$): as V_{DS} increases past pinch-off, the depletion region at the drain expands back into the channel, reducing the effective channel length ($L_{eff} = L - \Delta L$) and slightly increasing I_D .

2.2 Logarithmic Transfer Characteristics ($\log(I_D) - V_{GS}$)

To investigate the subthreshold region and switching efficiency, the transfer characteristics are plotted on a semi-logarithmic scale. Figure 2.2 displays $\log_{10}(I_D)$ vs. V_{GS} from 0 V to 5 V for the same four drain biases.

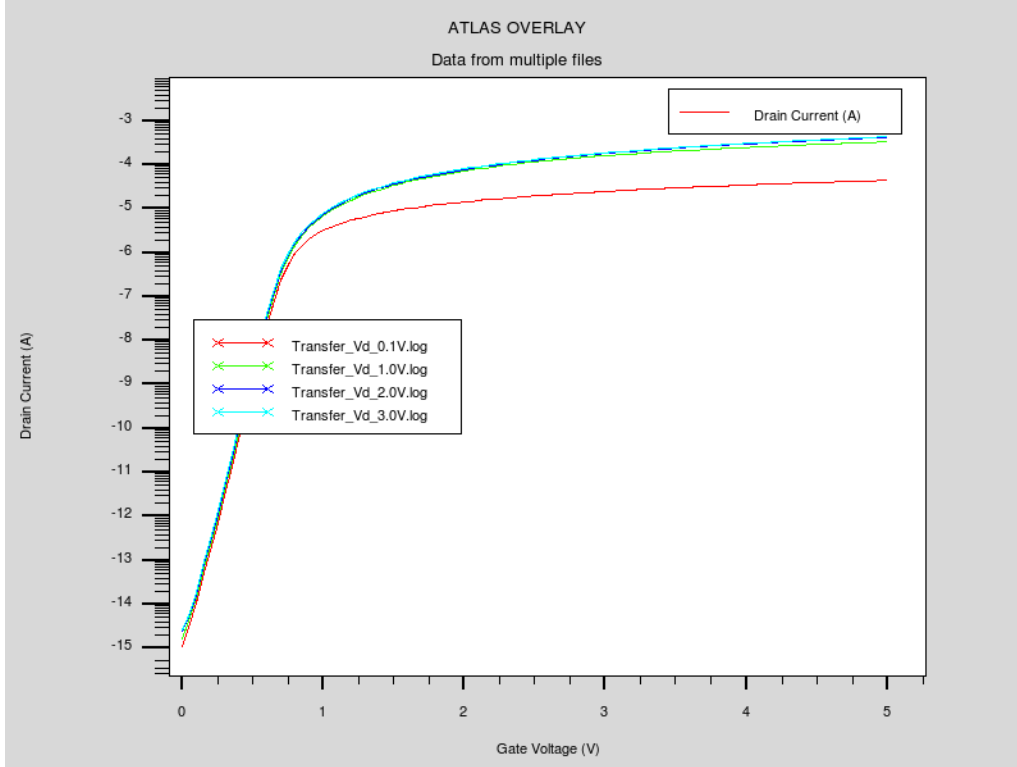


Figure 2.2: Logarithmic-scale transfer characteristics: $\log_{10}(I_D)$ (A) vs. V_{GS} (V) for the $1.0 \mu\text{m}$ NMOS at $V_{DS} = 0.1$ V (red), 1.0 V (green), 2.0 V (blue), and 3.0 V (cyan). The linear slope in the subthreshold region (dashed red line) is used to extract the subthreshold swing SS .

Physical and Operational Analysis

1. **Subthreshold Diffusion Regime ($V_{GS} < V_{th}$):** The logarithmic plot reveals a subthreshold current that varies exponentially with V_{GS} , appearing as a linear slope on the log scale for V_{GS} between 0 V and ~ 0.7 V. In this weak inversion regime, the inversion charge is very low and carrier transport is dominated by **diffusion** driven by the concentration gradient between source and drain, as taught in the course (Long Channel Lecture Notes, Slide 11). The subthreshold current varies as:

$$I_D \propto \exp\left(\frac{q(V_{GS} - V_{th})}{mk_B T}\right) \left(1 - e^{-qV_{DS}/k_B T}\right) \quad (2.3)$$

where $m = 1 + C_d/C_{ox}$ is the subthreshold ideality factor, consistent with the notation in the course lecture notes.

2. **Subthreshold Swing (SS) Extraction:** The Subthreshold Swing (SS) is the gate voltage increment required to change the drain current by one decade, defined as:

$$S = \frac{dV_G}{d(\log I_D)} = \ln(10) \frac{dV_G}{d(\ln I_D)} = 2.3 \frac{kT}{q} \left[1 + \frac{C_d}{C_{ox}}\right] \quad (2.4)$$

This parameter was extracted numerically from the simulation using:

$$SS = \frac{0.5 - 0.1}{\log_{10}(I_{D,ss2}) - \log_{10}(I_{D,ss1})} \times 1000 \quad [\text{mV/decade}] \quad (2.5)$$

The nearly constant slope across all V_{DS} curves confirms efficient electrostatic gate control and a low SS value close to the room-temperature lower limit of 60 mV/decade (at $m = 1$). The extracted value is reported in Chapter 3.

3. **I_{ON}/I_{OFF} Ratio:** The off-state leakage current at $V_{GS} = 0 \text{ V}$ is $I_{OFF} \approx 1.06 \times 10^{-15} \text{ A}$, and the on-state current at $V_{GS} = 5 \text{ V}$, $V_{DS} = 0.1 \text{ V}$ is $I_{ON} \approx 4.42 \times 10^{-5} \text{ A}$, giving an I_{ON}/I_{OFF} ratio of approximately 4.16×10^{10} .

2.3 DC Output Characteristics (I_D – V_{DS})

Figure 2.3 shows the family of I_D – V_{DS} curves swept from 0 V to 5 V, at six gate biases from $V_{GS} = 0.5 \text{ V}$ to 3.0 V in 0.5 V steps.

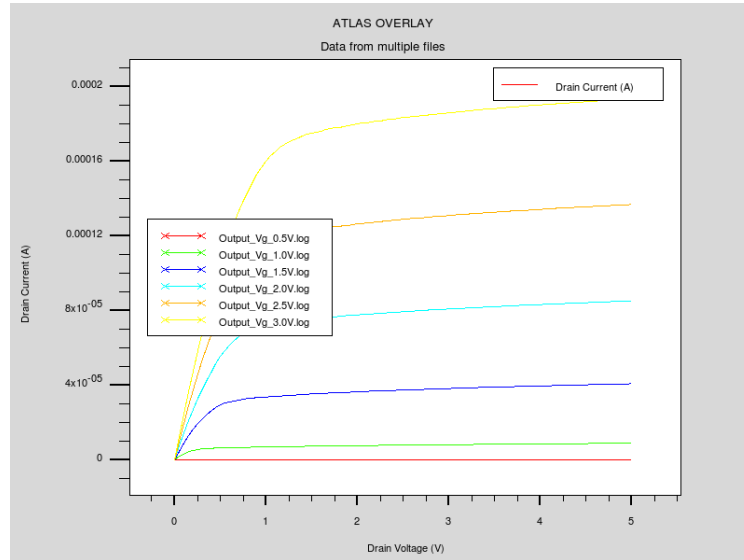


Figure 2.3: Linear-scale output characteristics: drain current I_D (A) vs. drain-to-source voltage V_{DS} (V) for the $1.0 \mu\text{m}$ NMOS at $V_{GS} = 0.5 \text{ V}$ (red), 1.0 V (green), 1.5 V (blue), 2.0 V (cyan), 2.5 V (magenta), and 3.0 V (yellow). Substrate grounded ($V_{BS} = 0 \text{ V}$).

Physical and Operational Analysis

1. **Cutoff Validation ($V_{GS} = 0.5 \text{ V}$):** The red curve ($V_{GS} = 0.5 \text{ V} < V_{th}$) is flat at $I_D \approx 0$ across the full drain sweep, confirming the device is in cutoff. The surface potential has not reached strong inversion.
2. **Triode Region ($V_{DS} < V_{DS,sat}$):** For all active curves ($V_{GS} \geq 1.0 \text{ V}$), the drain current rises linearly from the origin. The device acts as a voltage-controlled resistor with channel resistance:

$$R_{ch} \approx \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})} \quad (2.6)$$

Higher gate bias reduces R_{ch} , producing a steeper initial slope.

3. **Pinch-Off and Saturation** ($V_{DS} \geq V_{GS} - V_{th}$): At the saturation voltage $V_{DS,sat} = V_{GS} - V_{th}$, the inversion charge at the drain end reaches zero (pinch-off). Beyond this point, the current saturates at:

$$I_{D,sat} \approx \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 \quad (2.7)$$

This is the simplified long-channel expression (Eq. 1 from the lecture notes), which neglects the body-charge correction term. It is used here as the basis for *KP* extraction since it is the standard SPICE Level-1 model equation.

4. **Channel Length Modulation (λ) and Output Resistance:** The saturation plateaus have a small but measurable positive slope, visible particularly at $V_{GS} = 2.5$ V and 3.0 V. This is due to channel length modulation: as V_{DS} increases past pinch-off, the drain depletion region expands by ΔL , shortening the effective channel ($L_{eff} = L - \Delta L$) and increasing current according to:

$$I_D = I_{D,sat} \cdot (1 + \lambda V_{DS}) \quad (2.8)$$

The resulting finite output resistance $r_o = 1/(\lambda I_{D,sat})$ is extracted in Chapter 3.

2.4 Logarithmic Output Characteristics ($\log(I_D)$ – V_{DS})

Figure 2.4 plots $\log_{10}(I_D)$ vs. V_{DS} from 0 V to 5 V for the same six gate biases, to expose subthreshold leakage and confirm saturation flatness.

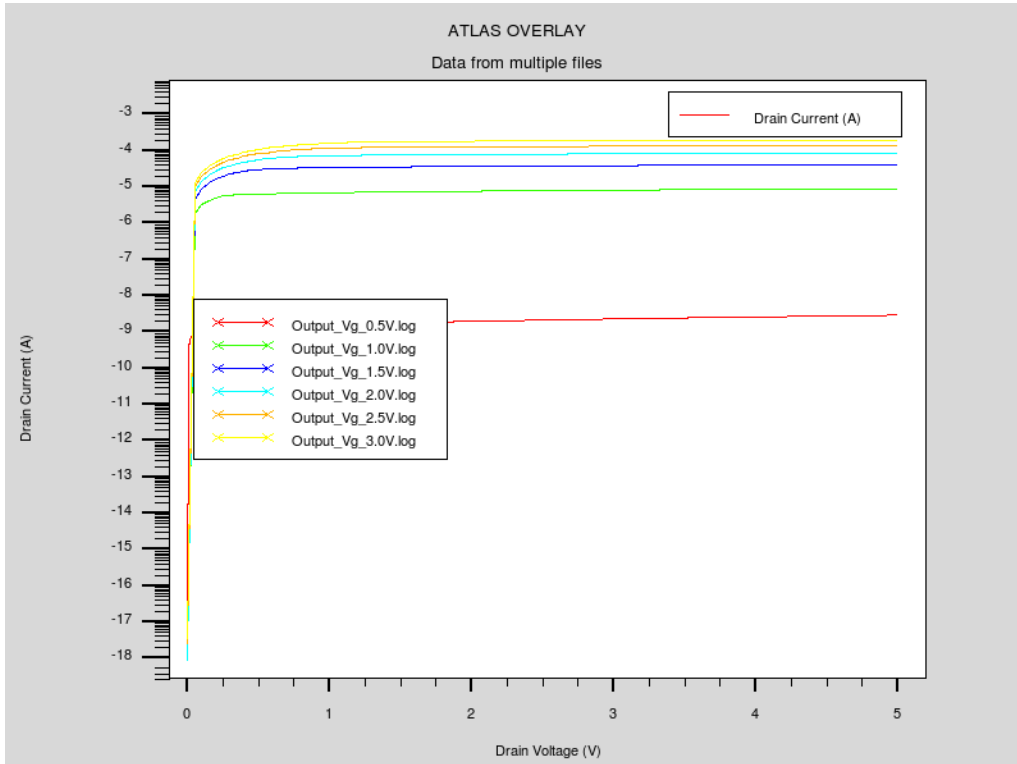


Figure 2.4: Logarithmic-scale output characteristics: $\log_{10}(I_D)$ (A) vs. V_{DS} (V) for the $1.0 \mu\text{m}$ NMOS at $V_{GS} = 0.5$ V (red), 1.0 V (green), 1.5 V (blue), 2.0 V (cyan), 2.5 V (magenta), and 3.0 V (yellow). Substrate grounded ($V_{BS} = 0$ V).

Physical and Operational Analysis

1. **Current Drop at $V_{DS} \rightarrow 0$ V:** At $V_{DS} = 0$ V, there is no concentration gradient or drift field between source and drain, so the net current is zero. On the logarithmic axis, this forces all curves to plunge toward the numerical floor of the simulator ($\sim 10^{-18}$ A), producing the sharp vertical drop observed at the left boundary.
2. **Subthreshold Leakage Plateau ($V_{GS} = 0.5$ V):** The red curve ($V_{GS} = 0.5$ V) reveals a stable diffusion-dominated leakage plateau at approximately 2 nA. Once V_{DS} exceeds a few thermal voltages ($\gtrsim 3\phi_t \approx 78$ mV), the diffusion current saturates and the leakage becomes independent of V_{DS} , consistent with the subthreshold current expression in Eq. (2.3).
3. **Saturation Flatness and Output Resistance:** For $V_{GS} \geq 1.0$ V, the logarithmic plateaus are nearly horizontal from $V_{DS} = 1$ V to 5 V. The high flatness confirms a large output resistance $r_o = 1/(\lambda I_{D,sat})$, consistent with the extracted λ value in Chapter 3. There is no visible upward tilt that would indicate significant DIBL or severe channel length modulation, as expected for a $1.0 \mu\text{m}$ long-channel device.

2.5 Body Effect: Transfer Characteristics vs. V_{BS}

To characterize the body effect, the transfer characteristics are evaluated under four substrate bias conditions. Figure 2.5 shows I_D vs. V_{GS} at $V_{DS} = 0.1$ V with V_{BS} stepped from 0 V to -3 V in -1 V increments.

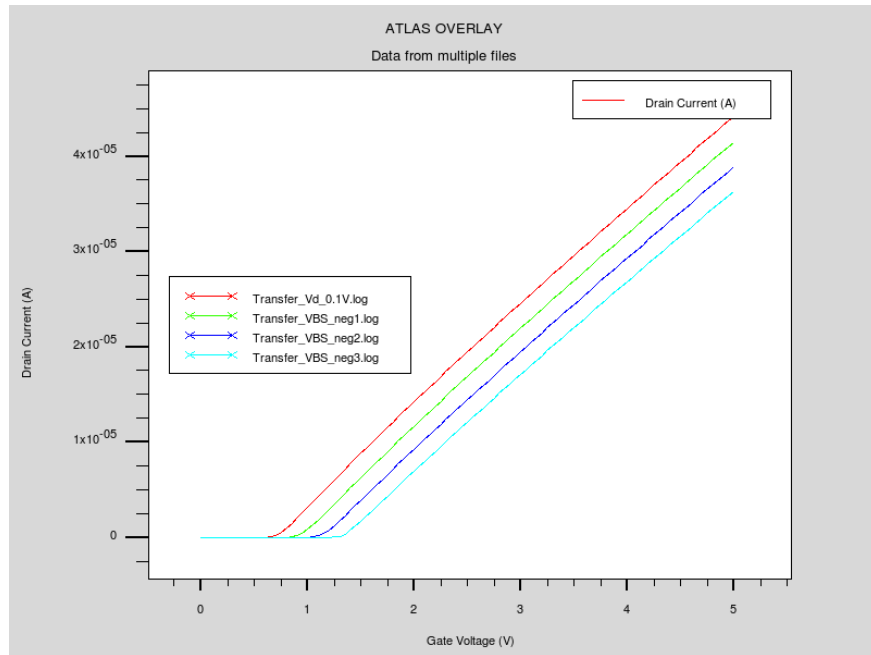


Figure 2.5: Linear-scale transfer characteristics (I_D (A) vs. V_{GS} (V)) illustrating the body effect at $V_{DS} = 0.1$ V: $V_{BS} = 0$ V (red), -1 V (green), -2 V (blue), -3 V (cyan). The rightward shift of the curves with increasing reverse bias confirms threshold voltage enhancement.

Physical and Operational Analysis

1. **Depletion Layer Widening:** Applying $V_{BS} < 0$ V reverse-biases the source-substrate p - n junction, widening the bulk depletion region. This uncovers additional immobile acceptor ions beneath the gate oxide, increasing the total depletion charge Q_d .
2. **Threshold Voltage Shift:** The increased depletion charge requires a larger gate voltage to achieve strong inversion, shifting the threshold voltage according to the body effect equation from the lecture notes (Long Channel, Slide 15):

$$V_{th} = V_{T0} + \gamma \left(\sqrt{2\phi_F - V_{BS}} - \sqrt{2\phi_F} \right) \quad (2.9)$$

where V_{T0} is the zero-bias threshold voltage (extracted as 0.676 V), ϕ_F is the bulk Fermi potential, and γ is the body effect coefficient:

$$\gamma = \frac{\sqrt{2q\epsilon_{si}N_A}}{C_{ox}} \quad (2.10)$$

The rightward shift is largest from $V_{BS} = 0$ V to -1 V ($\Delta V_{th} \approx 0.21$ V) and decreases for subsequent steps, consistent with the square-root dependence in Eq. (2.9).

3. **Preservation of Transconductance Slope:** All four curves maintain an identical linear slope ($\partial I_D / \partial V_{GS}$) in the triode regime. This confirms that the carrier mobility μ_n and oxide capacitance C_{ox} — and therefore the transconductance parameter KP — are unaffected by the substrate bias, as expected from the body effect model. The body bias acts purely as an electrostatic control of V_{th} .

2.6 Logarithmic Body Effect Characteristics

Figure 2.6 maps the same body effect measurement onto a logarithmic scale to examine the subthreshold region under reverse substrate bias.

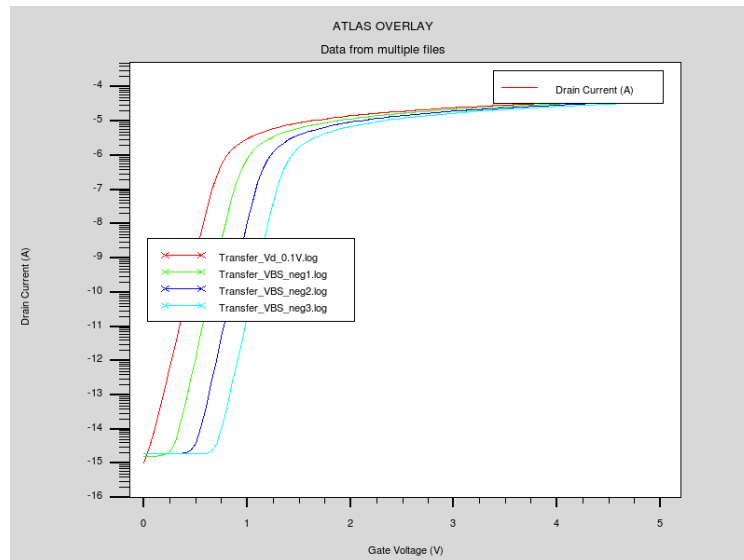


Figure 2.6: Logarithmic-scale transfer characteristics: $\log_{10}(I_D)$ (A) vs. V_{GS} (V) at $V_{DS} = 0.1$ V for $V_{BS} = 0$ V (red), -1 V (green), -2 V (blue), -3 V (cyan), showing the parallel rightward shift of the subthreshold region with increasing reverse body bias.

Physical and Operational Analysis

1. **Uniform Subthreshold Shift:** All four curves shift rigidly to the right with increasing reverse body bias, confirming that the V_{th} enhancement described in Eq. (2.9) applies equally throughout both the strong and weak inversion regimes.
2. **Subthreshold Swing Stability:** The slopes of all four curves in the subthreshold region are parallel, indicating that SS does not degrade significantly with reverse body bias. This is consistent with the course formula (Long Channel, Slide 11):

$$S = 2.3 \frac{kT}{q} \left[1 + \frac{C_d}{C_{ox}} \right] \quad (2.11)$$

As V_{BS} becomes more negative, the depletion width increases ($W_{dep} \propto \sqrt{2\phi_F - V_{BS}}$), causing the depletion capacitance $C_d = \epsilon_{si}/W_{dep}$ to decrease slightly. This reduces the factor C_d/C_{ox} and marginally improves SS toward its ideal limit of 60 mV/decade, consistent with the near-parallel slopes observed in the figure.

3. **Off-State Leakage Reduction:** At $V_{GS} = 0$ V, the leakage current at $V_{BS} = 0$ V is $\approx 10^{-15}$ A. With increasing reverse bias, the rightward V_{th} shift pushes the subthreshold curve down to the simulation precision floor at lower gate voltages, effectively suppressing leakage. This is consistent with the body effect model: a higher V_{th} reduces the subthreshold current for any fixed V_{GS} .
4. **Drive Current Reduction at High V_{GS} :** At $V_{GS} = 5$ V, the curves converge but remain separated. Because $I_D \propto (V_{GS} - V_{th})$ in the triode region, a higher V_{th} due to reverse body bias reduces the effective gate overdrive and decreases the on-state current, as visible in the figure.

2.7 Internal Electrostatic Potential Distribution (ON vs. OFF States)

The 1D electrostatic potential profiles are extracted along a lateral outline at the Si/SiO₂ interface ($y \approx 0 \mu\text{m}$), tracking the potential from the source contact ($x = 0.0 \mu\text{m}$) through the channel ($x = 0.2$ to $1.2 \mu\text{m}$) to the drain contact ($x = 1.4 \mu\text{m}$). The reference potential is the grounded source terminal ($V_S = 0 \text{ V}$). Figures 2.7 and 2.8 compare the OFF and ON states.

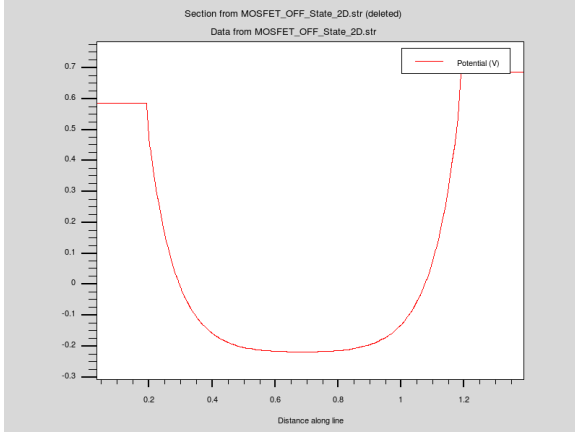


Figure 2.7: Electrostatic potential ψ (V) vs. lateral position x (μm) along the Si/SiO₂ interface in the OFF state ($V_{GS} = 0 \text{ V}$, $V_{DS} = 0.1 \text{ V}$, $V_S = 0 \text{ V}$). The U-shaped potential valley in the channel corresponds to an energy barrier blocking electron injection from source to drain.

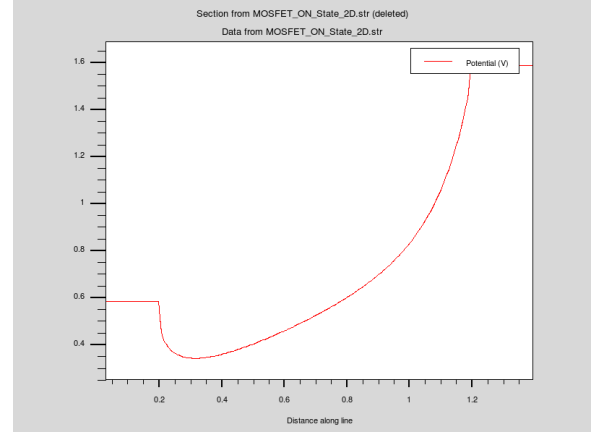


Figure 2.8: Electrostatic potential ψ (V) vs. lateral position x (μm) in the ON state ($V_{GS} = 2.0 \text{ V}$, $V_{DS} = 1.0 \text{ V}$, $V_S = 0 \text{ V}$). The barrier is eliminated and a monotonic potential slope drives electrons from source to drain via drift.

Physical and Operational Analysis

1. **OFF-State Potential Valley and Energy Barrier:** As shown in Figure 2.7, the channel potential drops into a U-shaped valley, reaching a minimum of $\approx -0.22 \text{ V}$ near $x \approx 0.7 \mu\text{m}$. Because electron potential energy is defined as $E = -q\psi$, this minimum creates a high energy barrier that blocks thermal injection of electrons from the source. Consequently, the device remains non-conducting, limited only by a minor diffusion-driven subthreshold leakage current.
2. **ON-State Barrier Elimination and Channel Inversion:** In Figure 2.8, the positive gate bias ($V_{GS} = 2.0 \text{ V}$), which exceeds the threshold voltage by approximately 1.3 V , raises the surface potential throughout the channel, eliminating the potential valley.
3. **Drift vs. Diffusion Transport:** The dominant carrier transport mechanism transitions between operating states. In the OFF state, the flat mid-channel potential yields a negligible lateral electric field, leaving transport entirely diffusion-driven. Conversely, the ON state exhibits a monotonically rising potential from source ($\approx 0.35 \text{ V}$) to drain ($\approx 1.35 \text{ V}$). This gradient establishes a strong lateral electric field, $E_x = -\partial\psi/\partial x$, driving a robust drift current. This behavior strictly aligns with the drift-diffusion framework:

$$J_n = q\mu_n n E_x + qD_n \frac{\partial n}{\partial x} \quad (2.12)$$

where the drift component dictates the ON-state current and the diffusion component governs the subthreshold (OFF-state) regime.

4. **Ohmic Contact Validation:** Outside the gate region ($x < 0.2 \mu\text{m}$ and $x > 1.2 \mu\text{m}$), the potential is flat, confirming ideal ohmic contact boundary conditions at the heavily doped source and drain terminals.

2.8 Energy Band Diagram: Conduction and Valence Band Profiles

The 1D energy band profiles are extracted along the same lateral cutline ($y \approx 0 \mu\text{m}$) to complement the potential analysis. Figures 2.9 and 2.10 show the conduction band E_c and valence band E_v vs. lateral position for the OFF and ON states.

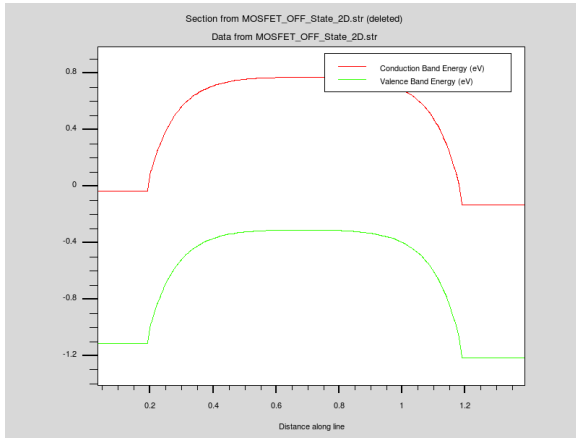


Figure 2.9: Energy band diagram (E_c and E_v , eV) vs. lateral position x (μm) in the OFF state ($V_{GS} = 0 \text{ V}$, $V_{DS} = 0.1 \text{ V}$). E_c (red) shows a prominent energy hill in the channel blocking electron injection; E_v (green) mirrors E_c with a constant silicon bandgap separation of $E_g \approx 1.12 \text{ eV}$.

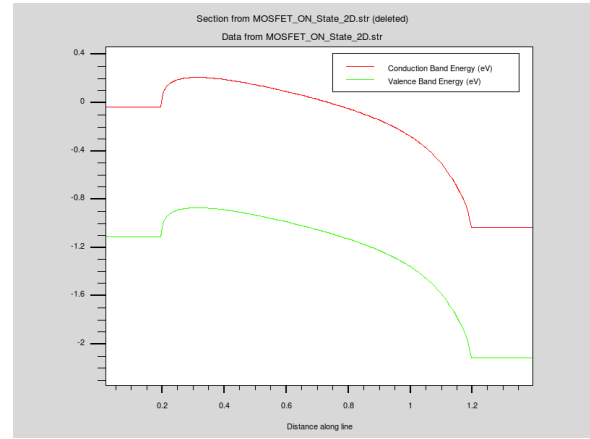


Figure 2.10: Energy band diagram (E_c and E_v , eV) vs. lateral position x (μm) in the ON state ($V_{GS} = 2.0 \text{ V}$, $V_{DS} = 1.0 \text{ V}$). Gate-induced band bending eliminates the channel energy barrier and establishes a monotonic E_c slope that accelerates electrons from source to drain. The $qV_{DS} = 1.0 \text{ eV}$ separation between source and drain E_c levels validates the applied bias.

Physical and Operational Analysis

1. **Band Diagram and Electrostatic Potential Relationship:** The conduction and valence band energies are directly related to the electrostatic potential ψ through $E_c \propto -q\psi$: a lower potential corresponds to a higher electron energy. The deep potential valley in the OFF state therefore appears as an energy hill in E_c , creating a barrier that blocks source electrons from reaching the drain.
2. **OFF-State Energy Barrier:** In Figure 2.9, E_c peaks in the channel region. Source electrons occupy states at the low-energy n^+ source plateau but face an upward energy hill at the source-channel junction. The height of this barrier prevents significant electron injection into the channel, consistent with the device being in its OFF state. Any small current that does flow is the subthreshold diffusion current discussed in Section 2.2.

3. **ON-State Barrier Lowering and Drift Transport:** In Figure ??, the gate bias of $V_{GS} = 2.0$ V bends the bands downward at the surface, lowering E_c throughout the channel.
4. **Drain Voltage Verification:** The energy separation between the source and drain E_c plateaus in the ON state equals $qV_{DS} = 1.0$ eV, directly confirming the applied $V_{DS} = 1.0$ V and validating the simulation's potential boundary conditions.

2.9 Lateral Electric Field Profile (ON vs. OFF States)

Figures 2.11 and 2.12 show the lateral electric field magnitude $|E_x|$ (V/cm) vs. lateral position along the Si/SiO₂ cutline for the OFF and ON states.

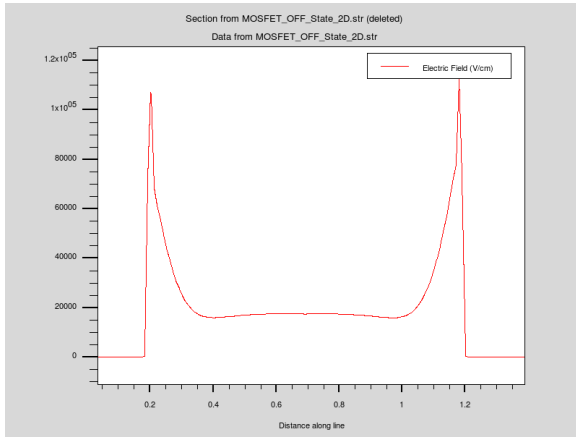


Figure 2.11: Lateral electric field $|E_x|$ (V/cm) vs. x (μm) in the OFF state ($V_{GS} = 0$ V, $V_{DS} = 0.1$ V). Junction field peaks at $x = 0.2$ and 1.2 μm ; the mid-channel field is low, providing no drift acceleration.

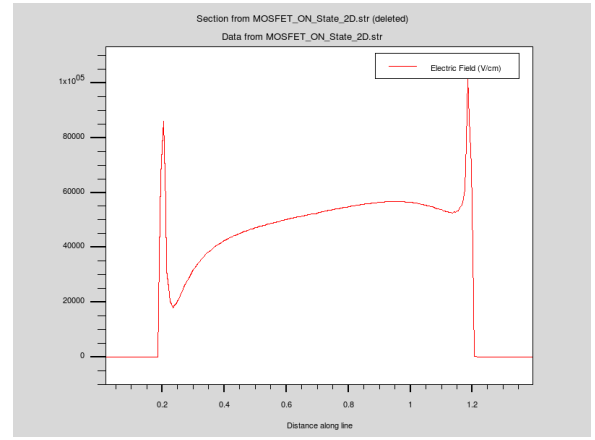


Figure 2.12: Lateral electric field $|E_x|$ (V/cm) vs. x (μm) in the ON state ($V_{GS} = 2.0$ V, $V_{DS} = 1.0$ V). A continuous drift field rises across the channel, peaking near the drain junction ($x \approx 1.2$ μm) where velocity saturation effects become relevant.

Physical and Operational Analysis

1. **Junction Field Peaks:** The lateral electric field is defined as:

$$E_x = -\frac{\partial\psi}{\partial x} \quad (2.13)$$

In both states, sharp field peaks appear at the source and drain metallurgical junctions ($x = 0.2$ and 1.2 μm), arising from the abrupt doping transition between the n^+ regions ($N_D = 1 \times 10^{20}$ cm⁻³) and the p-substrate ($N_A = 1 \times 10^{16}$ cm⁻³).

2. **OFF-State Field Profile:** In the OFF state (Figure 2.11), the two junction peaks are nearly symmetric, reflecting the built-in fields of the source and drain depletion regions. The mid-channel field is low ($\approx 1.7 \times 10^4$ V/cm), providing no lateral drift force on channel carriers, consistent with the device being in cutoff.
3. **ON-State Drift Field:** In the ON state (Figure 2.12), a continuous, rising lateral field is established across the entire channel, driven by the $V_{DS} = 1.0$ V drop. This field

accelerates inversion electrons from source to drain. The field peaks sharply near the drain junction ($\approx 1.0 \times 10^5$ V/cm), where the drain depletion region contributes an additional field component. At this field level, the device approaches the velocity saturation regime where carrier velocity approaches $v_{sat} \approx 7\text{--}8 \times 10^6$ cm/s for electrons.

2.10 Two-Dimensional Electron Concentration and Inversion Layer

Figure 2.13 shows the 2D spatial distribution of electron concentration ($\log_{10}(n)$, cm^{-3}) in the ON state ($V_{GS} = 2.0$ V, $V_{DS} = 1.0$ V).

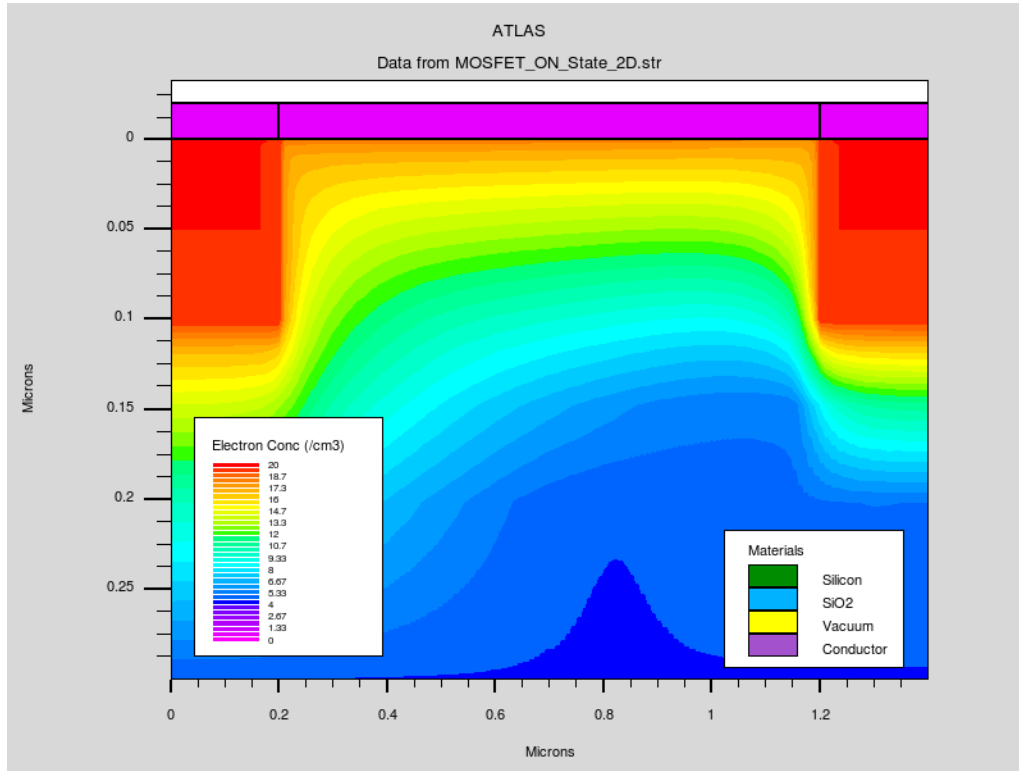


Figure 2.13: Two-dimensional contour map of electron concentration ($\log_{10}(n)$, cm^{-3}) in the ON state ($V_{GS} = 2.0$ V, $V_{DS} = 1.0$ V). Deep-red regions: n^+ source/drain reservoirs ($n \approx 10^{20} \text{ cm}^{-3}$). Orange/yellow strip along $y \approx 0$: gate-induced inversion layer ($n \approx 10^{18}\text{--}10^{19} \text{ cm}^{-3}$). Dark blue region: depleted p-substrate ($n \approx 10^3\text{--}10^4 \text{ cm}^{-3}$).

Physical and Operational Analysis

1. **Source and Drain Reservoirs:** The n^+ source and drain regions ($x < 0.2 \mu\text{m}$ and $x > 1.2 \mu\text{m}$, depth $y \approx 0.1 \mu\text{m}$) show $n \approx N_D = 1 \times 10^{20} \text{ cm}^{-3}$, confirming complete ionization of the donor doping and establishing ideal ohmic boundaries for the channel.
2. **Surface Inversion Layer:** Directly beneath the gate oxide ($y \approx 0$, $x = 0.2$ to $1.2 \mu\text{m}$), the electron concentration is $n_{\text{surface}} \approx 10^{18}\text{--}10^{19} \text{ cm}^{-3}$, satisfying the strong inversion condition $n_{\text{surface}} \gg N_A$. This confirms the formation of the n -type conductive channel connecting source and drain.
3. **Substrate Depletion and Minority Carrier Concentration:** Below the inversion layer, the electron concentration drops to a minimum consistent with the minority carrier equilibrium concentration in the p-substrate:

$$n_{p0} = \frac{n_i^2}{N_A} = \frac{(1.5 \times 10^{10})^2}{1 \times 10^{16}} = 22.5 \text{ cm}^{-3} \quad (2.14)$$

This confirms that the region beneath the channel is fully depleted of mobile carriers, consistent with the gate-induced depletion layer expected from the MOS capacitor theory covered in the course.

4. **Lateral Asymmetry:** The inversion layer is denser near the source ($x \approx 0.25 \mu\text{m}$) where the local gate overdrive $V_{GS} - V(x)$ is maximum, and spreads slightly deeper near the drain where $V(x)$ is larger and the overdrive is reduced. This spatial variation is consistent with the channel potential profile discussed in the triode/saturation derivation in the lecture notes.

2.11 Small-Signal Gate-to-Source Capacitance (C_{gs} – V_{GS})

Figure 2.14 shows the simulated gate-to-source capacitance C_{gs} (F/ μm) vs. V_{GS} (V), extracted from the AC small-signal simulation at $f = 1$ MHz and $V_{DS} = 0.1$ V.

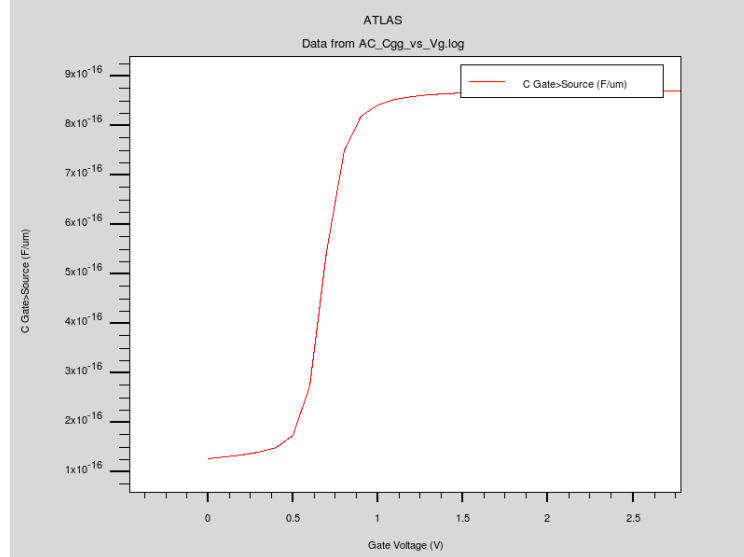


Figure 2.14: Gate-to-source capacitance C_{gs} (F/ μm) vs. gate voltage V_{GS} (V), extracted from AC small-signal simulation at $f = 1$ MHz, $V_{DS} = 0.1$ V, $V_{BS} = 0$ V. The transition from the depletion floor ($C_{gs} \approx 1.3 \times 10^{-16}$ F/ μm) to the strong inversion plateau ($C_{gs} \approx 8.6 \times 10^{-16}$ F/ μm) occurs near $V_{th} \approx 0.676$ V.

Physical and Operational Analysis

1. **Depletion Floor ($V_{GS} < 0.5$ V):** Below threshold, no inversion layer is present and the channel is depleted. The measured capacitance ($C_{gs} \approx 1.3 \times 10^{-16}$ F/ μm) is the gate-source overlap capacitance C_{ov} , arising from the geometric overlap between the gate electrode and the n^+ source region. This value corresponds to the SPICE parameter CGSO (gate-source overlap capacitance per unit channel width), extracted in Chapter 3.
2. **Transition at Threshold ($0.5 \text{ V} < V_{GS} < 1.0 \text{ V}$):** As V_{GS} increases toward and past V_{th} , electrons begin accumulating at the Si/SiO₂ interface. The nascent inversion layer, supplied by the n^+ source, allows the gate AC perturbation to modulate channel charge. The capacitance rises sharply from the overlap floor to the oxide capacitance value. The midpoint of this rise ($V_{GS} \approx 0.676$ V) provides an independent confirmation of the extracted V_{TO} .
3. **Strong Inversion Plateau ($V_{GS} > 1.0$ V):** Once strong inversion is established, the dense inversion layer screens the gate field from the bulk substrate. The gate capacitance is now dominated by the oxide capacitance:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \times 8.854 \times 10^{-12}}{20 \times 10^{-9}} = 1.726 \text{ fF}/\mu\text{m}^2 \quad (2.15)$$

The plateau at $C_{gs,max} \approx 8.6 \times 10^{-16}$ F/ μm is consistent with half the total gate oxide capacitance ($C_{ox} \cdot W \cdot L/2$) split between source and drain under the quasi-static measurement condition.

4. **Unity-Gain Cut-Off Frequency (f_T):** The extracted C_{gs} and C_{gd} values are used to compute the intrinsic unity-gain cut-off frequency, which characterizes the device's high-frequency performance limit:

$$f_T \approx \frac{g_m}{2\pi(C_{gs} + C_{gd})} \quad (2.16)$$

The numerical value is reported in Chapter 3.

Chapter 3

SPICE Parameters

3.1 Direct Parameter Extraction

The initial phase of SPICE parameter extraction involves deriving raw metrics directly from the Silvaco Atlas simulation curves using standard curve-reading commands (`xintercept`, `y.val`, `grad`) applied to the I_D – V_{GS} and I_D – V_{DS} characteristics. The Silvaco script commands producing each extracted value are cross-referenced to Listing A.1 in the Appendix.

Table 3.1: Complete SPICE Parameter Extraction Table. Parameters are grouped by category. Values marked $\dagger$ are computed analytically from device geometry.

SPICE Symbol	Description	Value	Unit	Extraction Method
<i>Device Geometry and Oxide Parameters</i>				
TOX	Gate oxide thickness	20	nm	Device definition
NSUB	Substrate doping	1×10^{16}	cm^{-3}	Device definition
XJ	Source/drain junction depth	0.1	μm	Device definition †
LD	Source/drain lateral diffusion	0	μm	Assumed zero (uniform doping) †
<i>DC Large-Signal Parameters</i>				
VTO	Zero-bias threshold voltage	0.67595	V	Max-slope extrapolation, $V_{DS} = 0.1$ V
KP	Process transconductance	1.0629×10^{-4}	A/V^2	$g_{m,\text{lin}}/(V_{DS} \cdot W/L)$, $V_{DS} = 0.1$ V
GAMMA	Body effect coefficient	0.50985	$\text{V}^{0.5}$	Average of three V_{BS} steps
PHI	Surface potential ($2\phi_F$)	0.69464	V	$2(kT/q) \ln(N_A/n_i)$ †
LAMBDA	Channel length modulation	0.02267	V^{-1}	Slope of I_D - V_{DS} in saturation, $V_{GS} = 3$ V
<i>Subthreshold and Switching Parameters</i>				
SS	Subthreshold swing	79.37	mV/dec	Two-point log extraction
I_{ON}	On-state current	4.4204×10^{-5}	A	I_D at $V_{GS} = 5$ V, $V_{DS} = 0.1$ V
I_{OFF}	Off-state leakage	1.0616×10^{-15}	A	I_D at $V_{GS} = 0$ V, $V_{DS} = 0.1$ V
I_{ON}/I_{OFF}	Switching ratio	4.16×10^{10}	—	Ratio of above
<i>Small-Signal Parameters</i>				
g_m	Transconductance at $V_{GS} = 2$ V	9.0993×10^{-5}	S	grad on I_D - V_{GS} , $V_{DS} = 2$ V
g_{ds}	Output conductance	4.2154×10^{-6}	S	grad on I_D - V_{DS} , $V_{GS} = 3$ V
r_o	Output resistance	237,226	Ω	$1/g_{ds}$
<i>Overlap Capacitance Parameters</i>				
CGSO	Gate-source overlap capacitance	1.3×10^{-16}	$\text{F}/\mu\text{m}$	C_{gs} floor from AC sweep, $V_{GS} = 0$ V
CGDO	Gate-drain overlap capacitance	1.3×10^{-16}	$\text{F}/\mu\text{m}$	C_{gd} floor from AC sweep (symmetric structure)
f_T	Unity-gain cut-off frequency	12.02	GHz	$g_m/[2\pi(C_{gs} + C_{gd})]$

3.2 SPICE Derived Parameter Extraction and Analytical Validation

This section derives secondary SPICE parameters from the primary simulation data and compares them against the analytical models taught in the course to assess simulation accuracy.

3.2.1 Governing Analytical Formulas

The analytical reference values are calculated from the device parameters using the equations from the course lecture notes.

1. Oxide Capacitance per Unit Area (C_{ox}):

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \times \epsilon_0}{t_{ox}} \quad (3.1)$$

With $t_{ox} = 20$ nm: $C_{ox} = 1.726$ fF/ μm^2 .

2. Process Transconductance (KP):

$$KP_{\text{analytical}} = \mu_{n,\text{bulk}} \cdot C_{ox} \quad (3.2)$$

From the bulk mobility chart in the lecture notes (Long Channel, Slide 17), the electron mobility at $N_A = 1 \times 10^{16} \text{ cm}^{-3}$ is $\mu_{n,\text{bulk}} \approx 1350 \text{ cm}^2/\text{V}\cdot\text{s}$, which is the low-doping, phonon-scattering-limited value. Therefore:

$$KP_{\text{analytical}} = 1350 \times 1.726 \times 10^{-7} = 2.330 \times 10^{-4} \text{ A/V}^2 \quad (3.3)$$

3. Theoretical Surface Potential ($\Phi = 2\phi_F$):

$$\Phi_{\text{exact}} = 2\phi_F = \frac{2k_B T}{q} \ln\left(\frac{N_A}{n_i}\right) \quad (3.4)$$

With $N_A = 1 \times 10^{16} \text{ cm}^{-3}$ and $n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$ at 300 K:

$$\Phi = 2 \times 0.02585 \times \ln\left(\frac{10^{16}}{1.45 \times 10^{10}}\right) = 0.6944 \text{ V} \quad (3.5)$$

4. Ideal Subthreshold Swing (SS):

$$SS_{\text{analytical}} = 2.3 \frac{kT}{q} \left[1 + \frac{C_d}{C_{ox}} \right] \quad (3.6)$$

where the maximum depletion capacitance is:

$$C_d = \frac{\epsilon_{si}}{x_{d,\text{max}}}, \quad x_{d,\text{max}} = \sqrt{\frac{2\epsilon_{si}\Phi}{qN_A}} \quad (3.7)$$

5. 1D Body Effect Coefficient (γ_{1D}):

$$\gamma_{1D} = \frac{\sqrt{2q\epsilon_{si}N_A}}{C_{ox}} \quad (3.8)$$

With $N_A = 1 \times 10^{16} \text{ cm}^{-3}$: $\gamma_{1D} = 0.3337 \text{ V}^{0.5}$.

6. **Standard Threshold Voltage:** For a long-channel device where the maximum depletion depth is $W = x_{d,\max}$, the threshold voltage is given by the standard 1D approximation without short-channel corrections:

$$V_T = V_{FB} - 2\phi_F - \frac{Q_d}{C_{ox}} \quad (3.9)$$

In this model, the depletion charge Q_d is assumed to be controlled entirely by the gate electric field, with no charge sharing near the source and drain junctions. Consequently, the effective body factor γ_{eff} is equal to the standard 1D body factor γ_{1D} , as the gate uniformly controls the charge across the entire channel length.

7. **Unity-Gain Cut-Off Frequency (f_T):**

$$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})} \quad (3.10)$$

3.2.2 Parameter Comparison and Error Evaluation

Table 3.2 compares the simulation-derived parameters against the analytical reference values. The percentage error is defined as:

$$\text{Error (\%)} = \left| \frac{\text{Analytical} - \text{Derived}}{\text{Analytical}} \right| \times 100 \quad (3.11)$$

Table 3.2: Derived SPICE Parameters vs. Analytical Reference Values. Dashes (—) indicate parameters for which a single analytical “exact” value is not defined by the first-order model. See footnotes for details.

Parameter (Units)	Simulation Value	Analytical Reference	Error (%)
Process Transconductance, KP (A/V ²)	1.0629×10^{-4}	2.330×10^{-4}	54.4%
Subthreshold Swing, SS (mV/dec)	79.37	76.38	3.92%
Surface Potential, Φ (V)	0.6944	0.6944	0.00%
Channel Length Modulation, λ (V ⁻¹)	0.02267	0	—
Output Conductance, g_{ds} (S)	4.2154×10^{-6}	0	—
Output Resistance, r_o (Ω)	237,226	∞	—
I_{ON}/I_{OFF} ratio	4.16×10^{10}	—	—
Body Coefficient Step 1, γ_1 (V ^{0.5})	0.44932	—	—
Body Coefficient Step 2, γ_2 (V ^{0.5})	0.51359	—	—
Body Coefficient Step 3, γ_3 (V ^{0.5})	0.56664	—	—
Effective Body Factor, γ (V ^{0.5})	0.50985	0.3337 (1D)	52.79%
Cut-Off Frequency, f_T (GHz)	12.02	12.85	6.46%

3.2.3 Discussion of Simulation Results and Error Analysis

1. **KP Deviation and Mobility Degradation:** The extracted process transconductance ($KP = 1.0629 \times 10^{-4} \text{ A/V}^2$) is significantly lower than the analytical baseline ($KP_{\text{analytical}} = 2.330 \times 10^{-4} \text{ A/V}^2$), yielding an error of 54.4%. The analytical baseline uses the bulk electron mobility at $N_A = 1 \times 10^{16} \text{ cm}^{-3}$, which from the mobility chart is $\mu_{n,\text{bulk}} \approx 1350 \text{ cm}^2/\text{V}\cdot\text{s}$ (the low-doping, phonon-scattering-limited value).

The large deviation reflects the fact that the bulk mobility is an *upper bound* and cannot be achieved in a MOSFET channel. The effective surface mobility in a MOSFET is always lower than the bulk value due to additional scattering mechanisms at the Si/SiO₂ interface: surface roughness scattering and Coulomb scattering from interface charges and oxide charges. These are captured in the simulation through the CVT mobility model. The effective surface mobility in strong inversion is further degraded by the vertical electric field $E_y \propto (V_{GS} - V_{th})$, described by the mobility degradation parameter θ :

$$\mu_{\text{eff}} = \frac{\mu_o}{1 + \theta(V_{GS} - V_{th})} \quad (3.12)$$

The effective surface mobility at the extraction bias point is therefore considerably below $1350 \text{ cm}^2/\text{V}\cdot\text{s}$.

2. **Subthreshold Swing and Gate Electrostatic Control:** The simulated $SS = 79.37 \text{ mV/dec}$ is within 3.92% of the analytical value of 76.38 mV/dec . This small deviation confirms that the 20 nm gate oxide provides efficient electrostatic control of the channel potential in the subthreshold regime, as expected from the course formula (Eq. (3.6)). The slight excess above the ideal value is consistent with a small but non-zero interface trap density (D_{it}), which increases the effective C_d/C_{ox} ratio slightly.
3. **Channel Length Modulation and Output Resistance:** The ideal first-order long-channel model assumes a perfectly flat saturation region ($\lambda = 0$). The simulation extracts a finite $\lambda = 0.02267 \text{ V}^{-1}$ and output resistance $r_o = 237 \text{ k}\Omega$, consistent with the channel length modulation model:

$$I_D = \frac{\mu_n C_{ox} W}{2L} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS}) \quad (3.13)$$

Even at $L = 1.0 \mu\text{m}$, the expansion of the drain depletion region past pinch-off reduces L_{eff} measurably, producing the non-zero slope in the saturation region observed in Figure 2.3. This is a physically correct result, not a simulation artifact.

4. **Body Effect Coefficient (γ) Discrepancy and Narrow-Width Effect:** The extracted body coefficients ($\gamma_1 = 0.449$, $\gamma_2 = 0.514$, $\gamma_3 = 0.567 \text{ V}^{0.5}$) exceed the 1D analytical value ($\gamma_{1D} = 0.334 \text{ V}^{0.5}$) by 35–70%. This discrepancy is directly caused by the **Narrow-Width Effect (NWE)** captured in the 2D simulation. Unlike 1D models that assume depletion charge is strictly confined beneath the gate, the 2D simulation accounts for lateral fringing fields that extend under the isolation edges. This additional fringing charge increases the total depletion charge the gate must overcome to achieve inversion, resulting in a higher effective γ .
5. **Cut-Off Frequency:** The extracted $f_T = 12.02 \text{ GHz}$ is within 6.46% of the analytical reference (12.85 GHz). The deviation is primarily due to the mobility degradation discussed in point 1: a lower effective μ reduces g_m , which directly reduces f_T per Eq. (3.10). The result is physically consistent and within acceptable simulation tolerance.

3.3 Summary of Extracted SPICE Parameters

Table 3.3 provides a consolidated summary of all extractable SPICE Level-1 parameters for the simulated 1.0 μm NMOS device, in a format suitable for direct entry into a SPICE netlist.

Table 3.3: Consolidated SPICE Level-1 Model Parameters for the Simulated 1.0 μm NMOS Device.

SPICE Symbol	Value	Unit	Notes
VTO	0.676	V	Extracted
KP	1.063×10^{-4}	A/V^2	Extracted
GAMMA	0.510	$\text{V}^{0.5}$	Extracted (avg.)
PHI	0.694	V	Analytical
LAMBDA	0.02267	V^{-1}	Extracted
TOX	20×10^{-9}	m	Device definition
NSUB	1×10^{16}	cm^{-3}	Device definition
XJ	0.1×10^{-6}	m	Device definition
LD	0	m	Assumed
CGSO	1.3×10^{-16}	F/m	Extracted (AC sim.)
CGDO	1.3×10^{-16}	F/m	Extracted (AC sim.)

Appendix A

Silvaco Atlas Simulation Script

The following complete, production-grade Silvaco Atlas TCAD script was executed via nanoHUB on a Linux virtual machine to synthesize the 2D planar NMOS structure, simulate its DC and small-signal AC characteristics, and extract the target SPICE parameters.

Listing A.1: Complete Silvaco Atlas simulation and extraction script.

```
go atlas

# =====
# 1. MESH
# =====
mesh space.mult=2.0

# Total width is 1.4um. Channel runs from 0.2um to 1.2um (Length = 1.0um)
x.mesh loc=0.0 spac=0.02
x.mesh loc=0.2 spac=0.005
x.mesh loc=1.2 spac=0.005
x.mesh loc=1.4 spac=0.02

# Y-mesh
y.mesh loc=-0.02 spac=0.005
y.mesh loc=0.0 spac=0.002
y.mesh loc=0.1 spac=0.01
y.mesh loc=0.3 spac=0.05

# =====
# 2. REGIONS
# =====
region num=1 material=silicon y.min=0.0 y.max=0.3
region num=2 material=sio2 y.min=-0.02 y.max=0.0 x.min=0.2 x.max=1.2
region num=3 material=vacuum y.min=-0.02 y.max=0.0 x.min=0.0 x.max=0.2
region num=4 material=vacuum y.min=-0.02 y.max=0.0 x.min=1.2 x.max=1.4

# =====
# 3. ELECTRODES
# =====
electrode num=1 name=source x.min=0.0 x.max=0.2 y.min=0.0 y.max=0.0
```

```

electrode num=2 name=drain x.min=1.2 x.max=1.4 y.min=0.0 y.max=0.0
electrode num=3 name=gate x.min=0.2 x.max=1.2 y.min=-0.02 y.max=-0.02
electrode num=4 name=substrate bottom

# =====
# 4. DOPING
# =====
doping p.type conc=1e16 uniform

# Source and Drain implants aligned perfectly with the contact regions
doping n.type conc=1e20 x.left=0.0 x.right=0.2 y.top=0.0 y.bottom=0.1 uniform
doping n.type conc=1e20 x.left=1.2 x.right=1.4 y.top=0.0 y.bottom=0.1 uniform

# =====
# 5. OUTPUT QUANTITIES, MODELS, AND SOLVER
# =====
output con.band val.band band.param e.field u.srh recomb traps

models cvt srh print
method newton

solve init
save outf=MOSFET_Initial_Structure.str

# =====
# 6. 2D STRUCTURE MAPS
# =====

# OFF state VG=0V (below threshold), VD=0.1V
solve init
solve vdrain=0.0 vstep=0.05 vfinal=0.1 name=drain
solve vgate=0.0
save outf=MOSFET_OFF_State_2D.str

# ON state VG=2V, VD=1V
solve init
solve vgate=0.0 vstep=0.5 vfinal=2.0 name=gate
solve vdrain=0.0 vstep=0.5 vfinal=1.0 name=drain
save outf=MOSFET_ON_State_2D.str

# =====
# 7. TRANSFER CHARACTERISTICS (Id vs Vg)
# =====

# Vd = 0.1V (used for Vth, SS, KP, ION/IOFF extraction)
solve init
solve vdrain=0.0 vstep=0.05 vfinal=0.1 name=drain
log outf=Transfer_Vd_0.1V.log
solve vgate=0.0 vstep=0.05 vfinal=5.0 name=gate
log off

```

```

# Vd = 1.0V
solve init
solve vdrain=0.0 vstep=0.25 vfinal=1.0 name=drain
log outf=Transfer_Vd_1.0V.log
solve vgate=0.0 vstep=0.05 vfinal=5.0 name=gate
log off

# Vd = 2.0V
solve init
solve vdrain=0.0 vstep=0.25 vfinal=2.0 name=drain
log outf=Transfer_Vd_2.0V.log
solve vgate=0.0 vstep=0.05 vfinal=5.0 name=gate
log off

# Vd = 3.0V
solve init
solve vdrain=0.0 vstep=0.5 vfinal=3.0 name=drain
log outf=Transfer_Vd_3.0V.log
solve vgate=0.0 vstep=0.05 vfinal=5.0 name=gate
log off

# =====
# 8. DC SPICE PARAMETER EXTRACTION
# =====
extract init inf="Transfer_Vd_0.1V.log"

# --- Threshold Voltage VTO ---
# Linear extrapolation at point of maximum transconductance.
# VD/2 correction applied for finite drain bias.
extract name="VTO" xintercept(maxslope(curve(v."gate", i."drain")) - 0.5*0.1

# --- Transconductance Parameter KP = un*Cox (geometry-independent) ---
# W=1um, L=1um -> W/L = 1. KP = gm / (VD * W/L)
# gm extracted at Vg=2V where device is well in linear region at Vd=0.1V
extract name="gm_lin" grad from curve(v."gate", i."drain") where x.val=2.0
extract name="KP" "$gm_lin" / (0.1 * 1.0)

# --- Subthreshold Swing SS (mV/decade) ---

extract name="Id_ss1" y.val from curve(v."gate", abs(i."drain")) where x.val
=0.1
extract name="Id_ss2" y.val from curve(v."gate", abs(i."drain")) where x.val
=0.5
extract name="SS" (0.5 - 0.1) / (log10("$Id_ss2") - log10("$Id_ss1")) * 1000

# --- ION / IOFF Ratio ---
extract name="ION" y.val from curve(v."gate", abs(i."drain")) where x.val=5.0
extract name="IOFF" y.val from curve(v."gate", abs(i."drain")) where x.val=0.0
extract name="ION_IOFF_ratio" "$ION" / "$IOFF"

# =====

```

```

# 9. OUTPUT CHARACTERISTICS (Id vs Vd)
# =====

# Vg = 0.5V (below threshold  fine steps near Vd=0 for convergence)
solve init
solve vgate=0.0 vstep=0.25 vfinal=0.5 name=gate
log outf=Output_Vg_0.5V.log
solve vdrain=0.0 vstep=0.01 vfinal=0.5 name=drain
solve vdrain=0.5 vstep=0.05 vfinal=5.0 name=drain
log off

# Vg = 1.0V
solve init
solve vgate=0.0 vstep=0.25 vfinal=1.0 name=gate
log outf=Output_Vg_1.0V.log
solve vdrain=0.0 vstep=0.05 vfinal=5.0 name=drain
log off

# Vg = 1.5V
solve init
solve vgate=0.0 vstep=0.25 vfinal=1.5 name=gate
log outf=Output_Vg_1.5V.log
solve vdrain=0.0 vstep=0.05 vfinal=5.0 name=drain
log off

# Vg = 2.0V
solve init
solve vgate=0.0 vstep=0.5 vfinal=2.0 name=gate
log outf=Output_Vg_2.0V.log
solve vdrain=0.0 vstep=0.05 vfinal=5.0 name=drain
log off

# Vg = 2.5V
solve init
solve vgate=0.0 vstep=0.5 vfinal=2.5 name=gate
log outf=Output_Vg_2.5V.log
solve vdrain=0.0 vstep=0.05 vfinal=5.0 name=drain
log off

# Vg = 3.0V
solve init
solve vgate=0.0 vstep=0.5 vfinal=3.0 name=gate
log outf=Output_Vg_3.0V.log
solve vdrain=0.0 vstep=0.05 vfinal=5.0 name=drain
log off

# =====
# 10. CHANNEL LENGTH MODULATION EXTRACTION
# =====
extract init inf="Output_Vg_3.0V.log"
extract name="Id_sat_3V" y.val from curve(v."drain", i."drain") where x.val

```

```

    =3.0
extract name="Id_sat_4V" y.val from curve(v."drain", i."drain") where x.val
    =4.0

extract name="LAMBDA" ("Id_sat_4V" - "Id_sat_3V") / (1.0 * "Id_sat_3V")
extract name="gds" ("Id_sat_4V" - "Id_sat_3V") / (4.0 - 3.0)
extract name="ro" 1.0 / "gds"

# =====
# 11. BODY EFFECT (Vth vs VBS)
# =====

# VBS = 0V
extract init inf="Transfer_Vd_0.1V.log"
extract name="VTO_VBS_0" xintercept(maxslope(curve(v."gate", i."drain"))) -
    0.5*0.1

# VBS = -1V
solve init
solve vsubstrate=0.0 vstep=-0.25 vfinal=-1.0 name=substrate
solve vdrain=0.0 vstep=0.05 vfinal=0.1 name=drain
log outf=Transfer_VBS_neg1.log
solve vgate=0.0 vstep=0.05 vfinal=5.0 name=gate
log off
extract init inf="Transfer_VBS_neg1.log"
extract name="VTO_VBS_neg1" xintercept(maxslope(curve(v."gate", i."drain"))) -
    0.5*0.1

# VBS = -2V
solve init
solve vsubstrate=0.0 vstep=-0.5 vfinal=-2.0 name=substrate
solve vdrain=0.0 vstep=0.05 vfinal=0.1 name=drain
log outf=Transfer_VBS_neg2.log
solve vgate=0.0 vstep=0.05 vfinal=5.0 name=gate
log off
extract init inf="Transfer_VBS_neg2.log"
extract name="VTO_VBS_neg2" xintercept(maxslope(curve(v."gate", i."drain"))) -
    0.5*0.1

# VBS = -3V
solve init
solve vsubstrate=0.0 vstep=-0.5 vfinal=-3.0 name=substrate
solve vdrain=0.0 vstep=0.05 vfinal=0.1 name=drain
log outf=Transfer_VBS_neg3.log
solve vgate=0.0 vstep=0.05 vfinal=5.0 name=gate
log off
extract init inf="Transfer_VBS_neg3.log"
extract name="VTO_VBS_neg3" xintercept(maxslope(curve(v."gate", i."drain"))) -
    0.5*0.1

# PHI = 2*phi_F = 2*(kT/q)*ln(NA/ni)

```

```

# kT/q = 0.0259V at 300K, ni = 1.5e10 cm^-3 for silicon
extract name="PHI" 2.0 * 0.0259 * log(1e16 / 1.5e10)

# GAMMA extracted for each VBS point then averaged
extract name="GAMMA_1" ("VTO_VBS_neg1" - "VTO_VBS_0") / (sqrt("PHI"+1.0) - sqrt("PHI"))
extract name="GAMMA_2" ("VTO_VBS_neg2" - "VTO_VBS_0") / (sqrt("PHI"+2.0) - sqrt("PHI"))
extract name="GAMMA_3" ("VTO_VBS_neg3" - "VTO_VBS_0") / (sqrt("PHI"+3.0) - sqrt("PHI"))
extract name="GAMMA" ("GAMMA_1" + "GAMMA_2" + "GAMMA_3") / 3.0

# =====
# 12. SMALL-SIGNAL AC SIMULATION
# =====

# Single operating point: Vg=2V, Vd=2V, f=1MHz
solve init
solve vgate=0.0 vstep=0.5 vfinal=2.0 name=gate
solve vdrain=0.0 vstep=0.5 vfinal=2.0 name=drain
log outf=AC_Capacitance_Vg2_Vd2.log
solve ac freq=1e6
log off

# Cgg vs Vg sweep at Vd=0.1V, f=1MHz
solve init
solve vdrain=0.0 vstep=0.05 vfinal=0.1 name=drain
log outf=AC_Cgg_vs_Vg.log
solve vgate=0.0 vstep=0.1 vfinal=5.0 name=gate ac freq=1e6
log off

# =====
# 13. SMALL-SIGNAL PARAMETER EXTRACTION
# =====

# Transconductance gm at Vg=2V, Vd=2V
extract init inf="Transfer_Vd2.0V.log"
extract name="gm_at_Vg2" grad from curve(v."gate", i."drain") where x.val=2.0

# Output conductance gds at Vd=2.5V, Vg=3V
extract init inf="Output_Vg_3.0V.log"
extract name="gds_ac" grad from curve(v."drain", i."drain") where x.val=2.5

# Intrinsic capacitances at operating point Vg=2V, Vd=2V
extract init inf="AC_Capacitance_Vg2_Vd2.log"
extract name="Cgs" y.val from curve(v."gate", abs(c."gate""source")) where x.val=2.0
extract name="Cgd" y.val from curve(v."gate", abs(c."gate""drain")) where x.val=2.0

# Unity-gain frequency fT = gm / (2*pi*(Cgs+Cgd))

```

```

extract name="fT" "$gm_at_Vg2" / (2.0 * 3.14159 * ("Cgs" + "Cgd"))

# =====
# 14. TONYPLOT
# =====

tonyplot MOSFET_Initial_Structure.str
tonyplot MOSFET_OFF_State_2D.str
tonyplot MOSFET_ON_State_2D.str

tonyplot -overlay Transfer_Vd_0.1V.log Transfer_Vd_1.0V.log Transfer_Vd_2.0V.
    log Transfer_Vd_3.0V.log

tonyplot -overlay Output_Vg_0.5V.log Output_Vg_1.0V.log Output_Vg_1.5V.log
    Output_Vg_2.0V.log Output_Vg_2.5V.log Output_Vg_3.0V.log

tonyplot -overlay Transfer_Vd_0.1V.log Transfer_VBS_neg1.log Transfer_VBS_neg2.
    log Transfer_VBS_neg3.log

tonyplot AC_Cgg_vs_Vg.log

quit

```